

Supplier Performance Model

Objective - Estimate the economic contribution of each supplier to the total cost of producing a reel, while accounting for differences in the grade being produced and the operating conditions.

$$C_i = \alpha + \delta_{\text{grade}(i)} + \sum_j \beta_j x_{i,j} + \sum_s \gamma_s p_{i,s} + \varepsilon_i$$

Reel cost = Baseline + Grade + Process + Supplier mix + Unexplained variation

What does each component mean?

C_i - Total cost of reel i

This is the cost that the model tries to explain. It includes the **production cost of the reel plus the cost of the raw material**, derived from the bales actually used to produce reel *i*. The objective is therefore broader than comparing supplier purchase prices: we want to understand contribution to the overall economic cost of production.

alpha - Baseline cost

This is the **intercept produced by the regression model once it has been fitted**. It provides the mathematical baseline from which the grade, process conditions and supplier composition increase or decrease the predicted cost.

delta grade(i) - Grade effect

Grade is represented as a **categorical variable** encoding the grade corresponding to reel *i*. The model therefore accounts for the fact that different grades can inherently have different production costs. This helps prevent a supplier from appearing more expensive simply because it is used more frequently for costly grades.

sum beta(j) x(i,j) - Process effect

Each $x_{i,j}$ is a process variable for reel *i* that is helpful for modelling cost - for example **machine speed, starch uptake, consistencies, temperatures**, and other relevant operating variables. For each process variable the regression estimates a coefficient β_j . Their combined contribution is:

$$\beta_1 x_{i,1} + \beta_2 x_{i,2} + \dots + \beta_J x_{i,J}$$

This part allows the model to account for differences in the conditions under which the reels were produced.

p(i,s) - Supplier proportion

A reel can be produced using bales from several suppliers. $p_{i,s}$ represents the **proportion of the raw material used for reel i that comes from supplier s**. The supplier proportions therefore describe the raw-material composition of each reel.

Supplier	Share of raw material
Supplier A	40%
Supplier B	35%
Supplier C	25%

gamma(s) - Supplier contribution

The regression estimates a coefficient γ_s associated with each supplier. This coefficient represents the supplier's estimated contribution to cost after accounting for the other factors represented in the model.

Lower supplier coefficient $\implies$ Lower estimated contribution to cost

The supplier coefficients can therefore provide the basis for an **adjusted economic comparison between suppliers**.

Different qualities from the same supplier

If the same supplier delivers significantly different bale qualities, these may need to be represented separately rather than assuming that all material from that supplier has the same economic effect. For example:

$$p_{\text{Supplier A, Quality 1}}, \quad p_{\text{Supplier A, Quality 2}}$$

The comparison could then be performed at the level of a **supplier-bale-quality combination**, with a separate estimated contribution for each relevant combination.

epsilon(i) - Unexplained variation

Not every difference in production cost can be explained by the information available to the model. ϵ_i represents the remaining unexplained variation for reel i .

What we ultimately want from the model

The purpose is not primarily to predict the cost of a reel. The important output is the estimated **supplier contribution**:

$$\gamma_1, \gamma_2, \dots, \gamma_S$$

These coefficients allow suppliers - or supplier/quality combinations - to be compared after accounting for grade and production conditions. A lower coefficient indicates a lower estimated contribution to total cost and therefore better economic performance according to the model.